

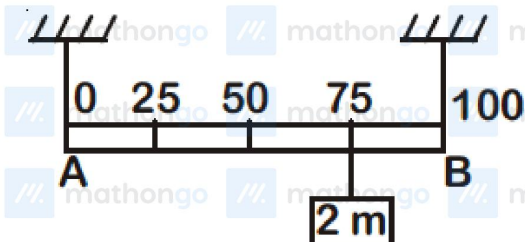
Rotation

Questions with Answer Keys

JEE Main 2020 Chapterwise

Physics

Q1 JEE Main 2020 - 2 September (Morning)

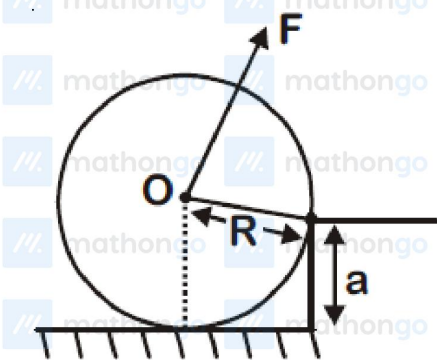


Shown in the figure is rigid and uniform one meter long rod AB held in horizontal position by two strings tied to its ends and attached to the ceiling. The rod is of mass ' m ' and has another weight of mass $2m$ hung at a distance of 75cm from A. The tension in the string at A is

- (A) $0.5mg$
- (B) $2mg$
- (C) $0.75mg$
- (D) $1mg$

Q2 JEE Main 2020 - 2 September (Morning)

A uniform cylinder of mass M and radius R is to be pulled over a step of height a ($a < R$) by applying a force F at its centre 'O' perpendicular to the plane through the axes of the cylinder on the edge of the step (see figure). The minimum value of F required is



- (A) $Mg \sqrt{1 - \left(\frac{R-a}{R}\right)^2}$

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(B) $Mg\sqrt{\left(\frac{R}{R-a}\right)^2 - 1}$

(C) $Mg\sqrt{1 - \frac{a^2}{R^2}}$

(D) $Mg\frac{a}{R}$

Q3 JEE Main 2020 - 2 September (Evening)

Two uniform circular discs are rotating independently in the same direction around their common axis passing through their centres. The moment of inertia and angular velocity of the first disc are $0.1\text{ kg} - \text{m}^2$ and 10 rad s^{-1} respectively while those for the second one are $0.2\text{ kg} - \text{m}^2$ and 5 rad s^{-1} respectively. At

some instant they get stuck together and start rotating as a single system about their common axis with some angular speed. The kinetic energy of the combined system is

(A) $\frac{20}{3}\text{ J}$

(B) $\frac{5}{3}\text{ J}$

(C) $\frac{10}{3}\text{ J}$

(D) $\frac{2}{3}\text{ J}$

Q4 JEE Main 2020 - 3 September (Morning)

Moment of inertia of a cylinder of mass M , length L and radius R about an axis passing through its centre and perpendicular to the axis of the cylinder is $I = M\left(\frac{R^2}{4} + \frac{L^2}{12}\right)$. If such a cylinder is to be made for a given mass of a material, the ratio L/R for it to have minimum possible I is

(A) $\sqrt{\frac{2}{3}}$

(B) $\frac{2}{3}$

(C) $\frac{3}{2}$

(D) $\sqrt{\frac{3}{2}}$

Q5 JEE Main 2020 - 3 September (Morning)

A block of mass $m = 1\text{ kg}$ slides with velocity $v = 6\text{ m/s}$ on a frictionless horizontal surface and collides with a uniform vertical rod and sticks to it as shown. The rod is pivoted about O and swings as

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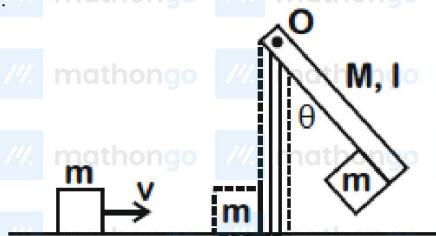
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a result of the collision making angle θ before momentarily coming to rest. If the rod has mass $M = 2\text{kg}$ and length $l = 1\text{m}$, the value of θ is approximately (take $g = 10\text{m/s}^2$)

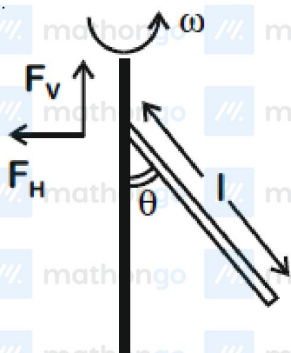


- (A) 49°
- (B) 55°
- (C) 63°
- (D) 69°

Q6 JEE Main 2020 - 3 September (Morning)

A person of 80kg mass is standing on the rim of a circular platform of mass 200kg rotating about its axis at 5 revolutions per minute (rpm). The person now starts moving towards the centre of the platform. What will be the rotational speed (in rpm) of the platform when the person reaches its centre

Q7 JEE Main 2020 - 3 September (Evening)



A uniform rod of length ' l ' is pivoted at one of its ends on a vertical shaft of negligible radius. When the shaft rotates at angular speed ω the rod makes an angle θ with it (see figure). To find θ equate the rate

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Physics

of change of angular momentum (direction going into the paper) $\frac{ml^2}{12}\omega^2 \sin \theta \cos \theta$ about the centre of mass (CM) to the torque provided by the horizontal and vertical forces F_H and F_v about the CM. The value of θ is then such that

(A) $\cos \theta = \frac{g}{|\omega^2|}$

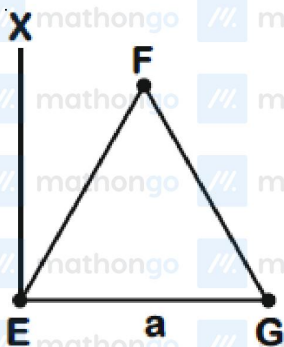
(B) $\cos \theta = \frac{2g}{3|\omega^2|}$

(C) $\cos \theta = \frac{g}{2|\omega^2|}$

(D) $\cos \theta = \frac{3g}{2|\omega^2|}$

Q8 JEE Main 2020 - 3 September (Evening)

An massless equilateral triangle EFG of side 'a' (As shown in figure) has three particles of mass m situated at its vertices. The moment of inertia of the system about the line EX perpendicular to EG in the plane of EFG is $\frac{N}{20}ma^2$ where N is an integer. The value of N is _____.



Q9 JEE Main 2020 - 4 September (Morning)

ABC is a plane lamina of the shape of an equilateral triangle. D, E are mid points of AB, AC and G is the centroid of the lamina. Moment of inertia of the lamina about an axis passing through G and perpendicular to the plane ABC is I_0 . If part ADE is removed, the moment of inertia of the remaining

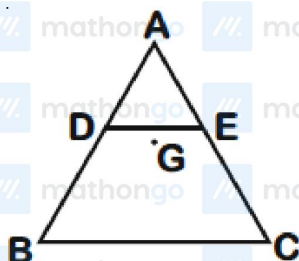
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part about the same axis is $\frac{NI_0}{16}$ where N is an integer. Value of N is _____.

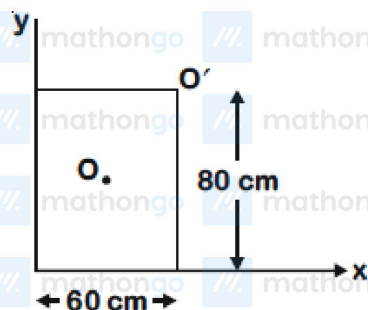


Q10 JEE Main 2020 - 4 September (Morning)

A circular disc of mass M and radius R is rotating about its axis with angular speed ω_1 . If another stationary disc having radius and same mass M is dropped co-axially on to the rotating disc. Gradually both discs attain constant angular speed ω_2 . The energy lost in the process is $p\%$ of the initial energy.

Value of p is

Q11 JEE Main 2020 - 4 September (Evening)



For a uniform rectangular sheet shown in the figure, the ratio of moments of inertia about the axes perpendicular to the sheet and passing through O (the centre of mass) and O' (corner point) is

- (A) $1/2$
- (B) $1/4$
- (C) $1/8$
- (D) $2/3$

Rotation

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Q12 JEE Main 2020 - 4 September (Evening)

Consider two uniform discs of the same thickness and different radii $R_1 = R$ and $R_2 = \alpha R$ made of the same material. If the ratio of their moments of inertia I_1 and I_2 , respectively, about their axes is

$I_1 : I_2 = 1 : 16$ then the value of α is

- (A) 2
- (B) 4
- (C) $2\sqrt{2}$
- (D) $\sqrt{2}$

Q13 JEE Main 2020 - 5 September (Morning)

A wheel is rotating freely with an angular speed ω on a shaft. The moment of inertia of the wheel is 1 and the moment of inertia of the shaft is negligible. Another wheel of moment of inertia $3i$ initially at rest is suddenly coupled to the same shaft. The resultant fractional loss in the kinetic energy of the system is

- (A) $\frac{5}{6}$
- (B) $\frac{1}{4}$
- (C) 0
- (D) $\frac{3}{4}$

Q14 JEE Main 2020 - 5 September (Morning)

A force $\vec{F} = (\hat{i} + 2\hat{j} + 3\hat{k})N$ acts at a point $(4\hat{i} + 3\hat{j} - \hat{k})m$. Then the magnitude of torque about the point $(\hat{i} + 2\hat{j} + \hat{k})m$ will be $\sqrt{x}N - m$. The value of x is

Q15 JEE Main 2020 - 5 September (Evening)

A thin rod of mass 0.9kg and length 1m is suspended, at rest, from one end so that it can freely oscillate in the vertical plane. A particle of mass 0.1kg moving in a straight line with velocity 80m/s hits the rod at its bottom most point and sticks to it (see figure). The angular speed (in rad/s) of the rod

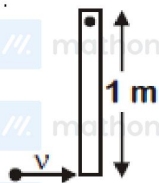
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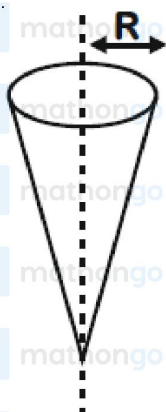
Physics

immediately after the collision will be



Q16 JEE Main 2020 - 6 September (Morning)

Shown in the figure is a hollow icecream cone (it open at the top). If its mass is M , radius of its top, R and height, H , then its moment of inertia about its axis is



- (A) $\frac{M(R^2 + H^2)}{4}$
(B) $\frac{MR^2}{2}$
(C) $\frac{MR^2}{3}$
(D) $\frac{MH^2}{3}$

Q17 JEE Main 2020 - 6 September (Morning)

Four point masses, each of mass m , are fixed at the corners of a square of side ℓ . The square is rotating with angular frequency ω , about an axis passing through one of the corners of the square and

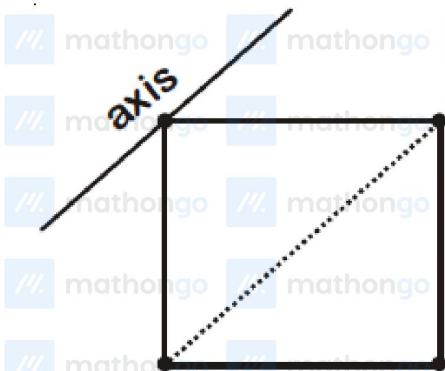
Rotation

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Physics

parallel to its diagonal, as shown in the figure. The angular momentum of the square about this axis is



- (A) $3m\ell^2\omega$
- (B) $4m\ell^2\omega$
- (C) $m\ell^2\omega$
- (D) $2m\ell^2\omega$

Q18 JEE Main 2020 - 6 September (Evening)

The linear mass density of a thin rod AB of length L varies from A to B as $\lambda(x) = \lambda_0 \left(1 + \frac{x}{L}\right)$, where x is the distance from A . If M is the mass of the rod then its moment of inertia about an axis passing through A and perpendicular to the rod is

- (A) $\frac{2}{5}ML^2$
- (B) $\frac{5}{12}ML^2$
- (C) $\frac{3}{7}ML^2$
- (D) $\frac{7}{18}ML^2$

Q19 JEE Main 2020 - 7 January (Morning)

The radius of gyration of a uniform rod of length L , about an axis passing through a point $\frac{L}{4}$ away from the centre of the rod, and perpendicular to it, is

- (A) $\sqrt{\frac{7}{48}}L$
- (B) $\sqrt{\frac{5}{48}}L$

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Physics

(C) $\sqrt{\frac{7}{24}}L$

(D) $\sqrt{\frac{19}{24}}L$

Q20 JEE Main 2020 - 7 January (Morning)

A satellite of mass m is launched vertically upwards with an initial speed u from the surface of the earth. After it reaches height R (R = radius of the earth), it ejects a rocket of mass $\frac{m}{10}$ so that subsequently the satellite moves in a circular orbit. The kinetic energy of the rocket is (G is the gravitational constant; M is the mass of the earth) :

(A) $5M \left(u^2 - \frac{119GM}{200R} \right)$

(B) $5M \left(u^2 - \frac{113GM}{200R} \right)$

(C) $\frac{M}{20} \left(u^2 - \frac{119GM}{100R} \right)$

(D) $\frac{M}{20} \left(u^2 - \frac{113GM}{200R} \right)$

Q21 JEE Main 2020 - 7 January (Evening)

Mass per unit area of a circular disc of radius a depends on the distance r from its centre as $\sigma(r) = A + Br$. The moment of inertia of the disc about the axis, perpendicular to the plane and passing through its centre is :

(A) $2\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$

(B) $2\pi a^4 \left(\frac{Aa}{4} + \frac{B}{5} \right)$

(C) $\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$

(D) $2\pi a^4 \left(\frac{A}{5} + \frac{Ba}{4} \right)$

Q22 JEE Main 2020 - 8 January (Morning)

Consider a uniform rod of mass $M = 4m$ and length L pivoted about its centre. A mass m moving with velocity v making angle $\theta = 4\pi$ to the rod's long axis collides with one end of the rod and sticks to it. The angular speed of the rod mass system just after the collision is:

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Rotation

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Physics

(A) $\frac{3\sqrt{2}V}{7L}$

(B) $\frac{\sqrt{2}V}{7L}$

(C) $\frac{\sqrt{2}V}{3L}$

(D) $\frac{3V}{7L}$

Q23 JEE Main 2020 - 8 January (Evening)

A solid sphere of mass $m = 500 \text{ gm}$ is rolling without slipping on a horizontal surface. Find kinetic energy of a sphere if velocity of centre of mass is 5 cm/sec .

(A) $\frac{35}{2} \times 10^{-4} \text{ J}$

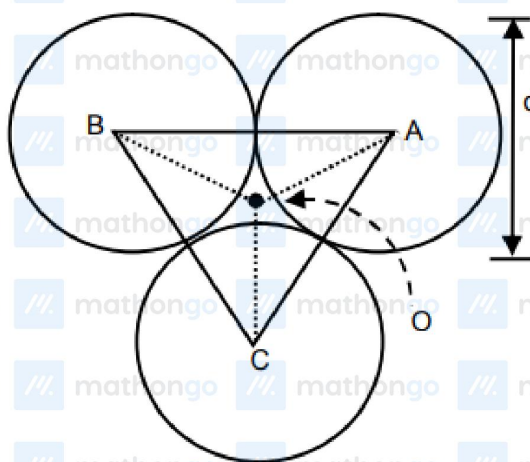
(B) $\frac{35}{4} \times 10^{-4} \text{ J}$

(C) $35 \times 10^{-4} \text{ J}$

(D) $35 \times 10^{-3} \text{ J}$

Q24 JEE Main 2020 - 9 January (Morning)

Three solid spheres each of mass m and diameter d are stuck together such that the lines connecting the centres form an equilateral triangle of side of length d . The ratio $\frac{I_o}{I_A}$ of moment of inertial I_o of the system about an axis passing the centroid and about center of any of the spheres I_A and perpendicular



to the plane of the triangle is :

(A) $13/23$

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Physics

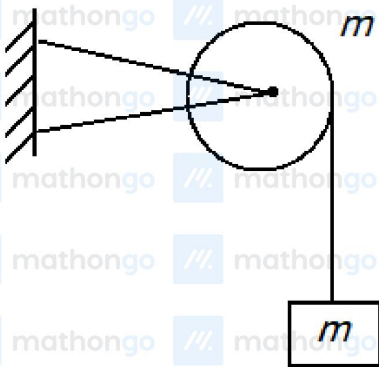
(B) $11/19$

(C) $7/9$

(D) $13/11$

Q25 JEE Main 2020 - 7 January (Morning)

As shown in the figure, a bob of mass m is tied by a massless string whose other end portion is wound on a fly wheel (disc) of radius R and mass m . When released from rest the bob starts falling vertically. When it has covered a distance of h , the angular speed of the wheel will be:



(A) $\frac{1}{R} \sqrt{\frac{4gh}{3}}$

(B) $\frac{1}{R} \sqrt{\frac{2gh}{3}}$

(C) $R \sqrt{\frac{2gh}{3}}$

(D) $R \sqrt{\frac{4gh}{3}}$

Rotation

Questions with Answer Keys

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Physics

Answer Key

Q1 (D)

Q2 (A)

Q3 (A)

Q4 (D)

Q5 (C)

Q6 (9)

Q7 (D)

Q8 (25)

Q9 (11)

Q10 (20)

Q11 (B)

Q12 (A)

Q13 (D)

Q14 (195)

Q15 (20)

Q16 (B)

Q17 (A)

Q18 (D)

Q19 (A)

Q20 (A)

Q21 (A)

Q22 (A)

Q23 (B)

Q24 (A)

Q25 (A)

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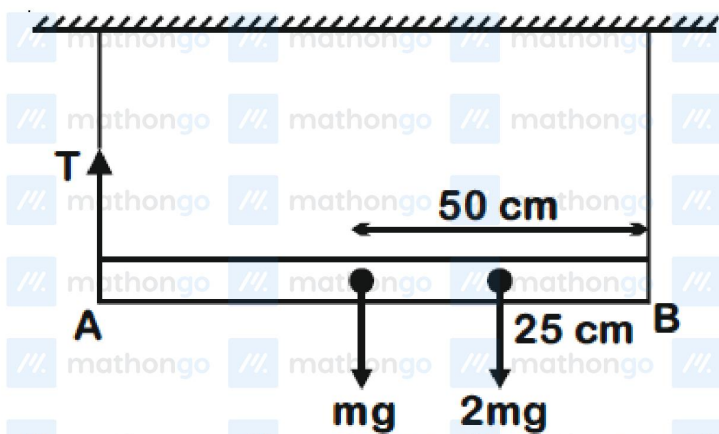
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Q1

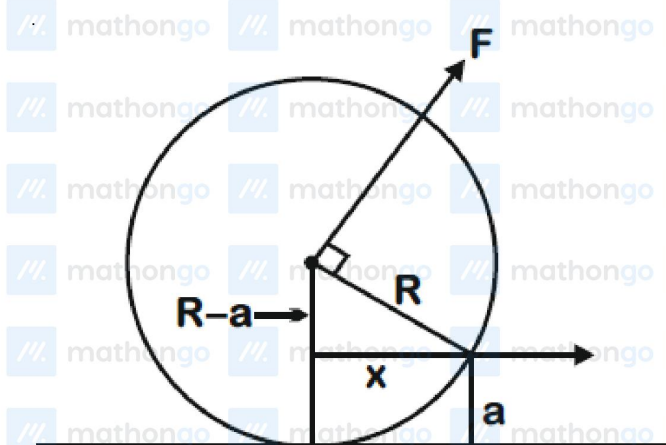


Net torque about $B = 0$

$$\Rightarrow T \times 100 = mg \times 50 + 2mg \times 25$$

$$\Rightarrow T = mg$$

Q2



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To Step up,

$$\begin{aligned} F \times R &\geq Mg \times x \\ \Rightarrow F_{\min} &= \frac{Mg}{R} \times \sqrt{R^2 - (R - a)^2} \\ &= Mg \sqrt{1 - \left(\frac{R - a}{R}\right)^2} \end{aligned}$$

Q3

$$(1_1 + 1_2) \omega = 1_1 \omega_1 + 1_2 \omega_2$$

$$\begin{aligned} \omega &= \frac{0.1 \times 10 + 0.2 \times 5}{0.1 + 0.2} \\ &= \frac{20}{3} \text{ rad/s} \end{aligned}$$

$$\begin{aligned} \text{K.E.} &= \frac{1}{2} (I_1 + I_2) \omega^2 \\ &= \frac{20}{3} \text{ J} \end{aligned}$$

Q4

$$I = M \left[\frac{R^2}{4} + \frac{L^2}{12} \right]$$

$$M = \rho \pi R^2 L$$

$$\therefore I = M \left[\frac{M}{4\rho\pi L} + \frac{L^2}{12} \right]$$

$$\therefore \frac{dI}{dL} = 0 \Rightarrow L^3 = \frac{6}{4} \frac{M}{\rho\pi}$$

$$\therefore \frac{L^3}{LR^2} = \frac{6}{4}$$

$$\Rightarrow \frac{L}{R} = \sqrt{\frac{3}{2}}$$

Q5

$$mvl = \left(\frac{MI^2}{3} + ml^2 \right) \omega$$

$$\frac{1}{2} \left(\frac{MI^2}{3} + ml^2 \right) \omega^2 = \left[Mg \frac{1}{2} + mg \right] (1 - \cos \theta)$$

$$\omega = \frac{18}{5} \text{ rad/s}$$

Rotation

Hints & Solutions

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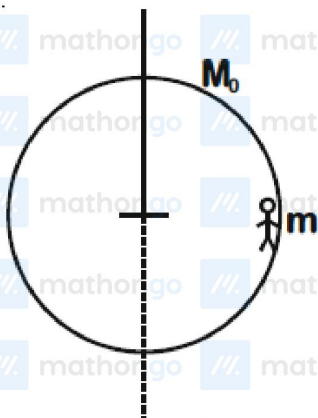
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$$\frac{1}{2} \left(\frac{5}{3} \right) \times \frac{324}{25} = (10 + 10)(1 - \cos \theta)$$

$$20(1 - \cos \theta) = \frac{324}{30}$$

$$\cos \theta = 1 - \frac{324}{600} = \frac{276}{600} = 0.46$$

Q6



$$M_0 = 200\text{kg}$$

$$m = 80\text{kg}$$

$$\therefore W_2 = \left(\frac{M_0 R^2}{2} + m R^2 \right) \times \frac{5}{\frac{M_0 R^2}{2}}$$

$$= \frac{5R^2}{R^2} \times \frac{(80+100)}{100}$$

$$W_2 = 9\text{RPM}$$

Q7

Rotation

Hints & Solutions

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Physics



From a rotating frame rod will appear in equilibrium. Net torque about suspension point must be zero.

$$\int_0^L \left(\frac{M}{L} dx \right) (x \sin \theta) \omega^2 x \cos \theta = Mg \frac{L}{2} \sin \theta$$

$$\Rightarrow \cos \theta = \frac{3g}{2|\omega|^2}$$

Q8

$$x_1 = \frac{a}{2}; x_2 = a$$

and $x_3 = 0$



$$I = \frac{ma^2}{4} + ma^2 + 0$$

$$= \frac{5}{4}ma^2$$

$$I = \frac{25}{20}ma^2 = \frac{N}{20}ma^2$$

$$\therefore N = 25$$

Q9

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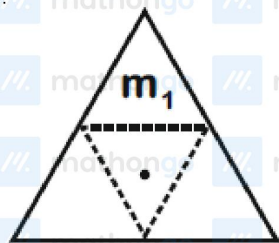
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Let I_0 is about 'G' mass be m , and length be a . $\therefore kma^2 = I_0 \dots (i)$

Then MI of $(m_1) = \frac{km}{4} \frac{a^2}{4} = \frac{I_0}{16}$

Let moment of inertia of one triangular part (side) be I_1 about 'G' then

$$3I_1 + \frac{I_0}{16} = I_0 \Rightarrow 3I_1 = I_0 - \frac{I_0}{16} = \frac{15I_0}{16}$$

$$\Rightarrow I_1 = \frac{5I_0}{16}$$

So moment of inertia of remaining portion $I'' = I_0 - \frac{5I_0}{16}$

$$\Rightarrow I'' = \frac{11}{16} I_0$$

Q10

$$I_1 \omega_1 = I_2 \omega_2$$

$$\omega_2 = \frac{\left(\frac{MR^2}{2}\right) \omega_1}{\left(\frac{MR^2}{2} + \frac{MR^2}{8}\right)} = \frac{MR^2}{2} \frac{\omega_1 \times 8}{MR^2 \times 5} = \frac{4\omega_1}{5}$$

$$\therefore \left(\frac{\Delta E}{E} \times 100\right) = P = \left(\frac{\frac{1}{2} \cdot \frac{MR^2}{2} \cdot \omega_1^2 - \frac{1}{2} \cdot \frac{5MR^2}{8} \cdot \frac{16}{25} \omega_1^2}{\frac{1}{2} \cdot \frac{MR^2}{2} \cdot \omega_1^2}\right) \times 100$$

$$= \frac{1}{2} \frac{MR^2}{2} \omega_1^2 \frac{\left(1 - \frac{4}{5}\right)}{\frac{1}{2} \frac{MR^2}{2} \cdot \omega_1^2} \times 100 = 20$$

Q11

Rotation

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$$I_0 = \frac{M}{12} ((80)^2 + (60)^2)$$

$$I_0 = I_0 + M(50)^2$$

$$\Rightarrow \frac{I_0}{I_0} = \frac{\frac{M}{12}(10000)}{\frac{M}{12}(10000) + (M)(2500)} = \frac{1}{4}$$

Q12

$$1 = MR^2$$

$$M \propto R^2$$

$$1 \propto R^4$$

$$\frac{L_1}{I_2} = \left(\frac{R_1}{R_2} \right)^4$$

Q13

$$I\omega = 4I\omega'$$

$$\Rightarrow \omega' = \frac{\omega}{4}$$

$$(\Delta K)_{\text{Loss}} = \frac{1}{2} I \omega^2 - \frac{1}{2} (4I) \left(\frac{\omega}{4} \right)^2$$

$$\Rightarrow \text{Fractional loss} = \frac{(\Delta K)_{\text{Loss}}}{K_1} \quad (K_1 = \frac{1}{2} I \omega^2)$$

$$= \frac{3}{4}$$

Q14

$$\vec{\tau} = \vec{r} \times \vec{f}$$

$$\vec{r} = (4\hat{i} + 3\hat{j} - \hat{k}) - (\hat{i} + 2\hat{j} + \hat{k})$$

$$\Rightarrow \vec{r} = (3\hat{i} + \hat{j} - 2\hat{k})$$

$$\vec{\tau} = (3\hat{i} + \hat{j} - 2\hat{k}) \times (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{\tau} = 6\hat{k} - 9\hat{j} - \hat{k} + 3\hat{i} - 2\hat{i} + 4\hat{i}$$

$$\vec{\tau} = 7\hat{i} - 11\hat{j} + 5\hat{k}$$

$$\therefore (\tau) = \sqrt{195}$$

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Q15

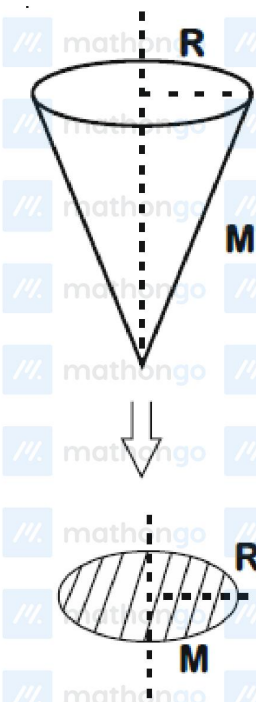


Angular momentum will remain conserved about pivotal point

$$(0.1 \times 80 \times 1) = \left[\left(\frac{1}{10} \times 1 \right) + \left(\frac{9}{10} \times \frac{1}{3} \right) \right] \omega$$

$$\Rightarrow \omega = \frac{8 \times 10}{4} = 20 \text{ rad/s}$$

Q16



$$\text{Sol. : } 1 = \frac{MR^2}{2}$$

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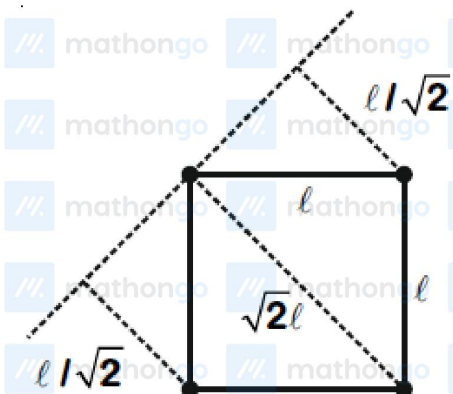
Hints & Solutions

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Moment of inertia of this cone will same as circular disk of mass (M) and radius R.

Q17



$$I = m \left(\frac{\ell^2}{2} \right) \times 2 + m \times (\sqrt{2}\ell)^2 = 3m\ell^2$$
$$\therefore L = I_W = 3m\ell^2\omega$$

Q18

$$M = \int_0^L \lambda dx = \int_0^L \lambda_0 \left(1 + \frac{x}{L} \right) dx$$
$$= \frac{3}{2} \lambda_0 L$$
$$I = \int_0^L (\lambda dx) x^2 = \frac{7}{12} \lambda_0 L^3$$
$$I = \frac{7}{18} ML^2$$

Q19

Rotation

Hints & Solutions

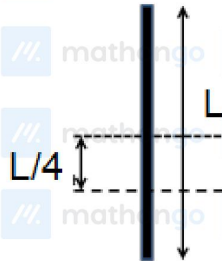
JEE Main 2020 Chapterwise

Physics

$$\frac{ML^2}{12} + M\left(\frac{L}{4}\right)^2 = MK^2$$

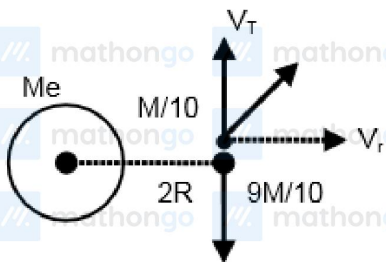
$$\frac{L^2}{12} + \frac{L^2}{16} = K^2$$

$$K = \sqrt{\frac{7}{48}}L$$



Q20

$$\frac{-GM_e M}{R} + \frac{1}{2}Mu^2 = \frac{-GM_e M}{2R} + \frac{1}{2}Mv^2$$



$V_T \rightarrow$ Transverse velocity of rocket
 $V_R \rightarrow$ Radial velocity of rocket

Rotation

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$v = \sqrt{u^2 - \frac{GM_e}{R}}$$

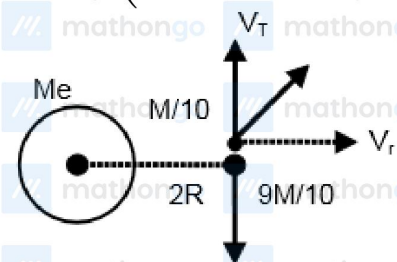
$$\frac{M}{10} V_T = \frac{9M}{10} \sqrt{\frac{GM_e}{2R}}$$

$$\frac{M}{10} V_r = M \sqrt{u^2 - \frac{GM_e}{R}}$$

$$\text{Kinetic energy} = \frac{1}{2} \frac{M}{10} (V_T^2 + V_r^2)$$

$$= \frac{M}{20} \left(81 \frac{GM_e}{2R} + 100u^2 - 100 \frac{GM_e}{R} \right)$$

$$= \frac{M}{20} \left(100u^2 - \frac{119GM_e}{2R} \right)$$



$V_T \rightarrow$ Transverse velocity of rocket

$V_r \rightarrow$ Radial velocity of rocket

Q21

Rotation

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$\sigma = A + Br$$

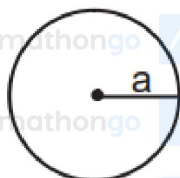
$$\int dm = \int (A + Br) 2\pi r dr$$

$$I = \int dm r^2$$

$$= \int_0^a (A + Br) 2\pi r^3 dr$$

$$= 2\pi \left(A \frac{a^4}{4} + B \frac{a^5}{5} \right)$$

$$= 2\pi a^4 \left(\frac{A}{4} + \frac{Ba}{5} \right)$$



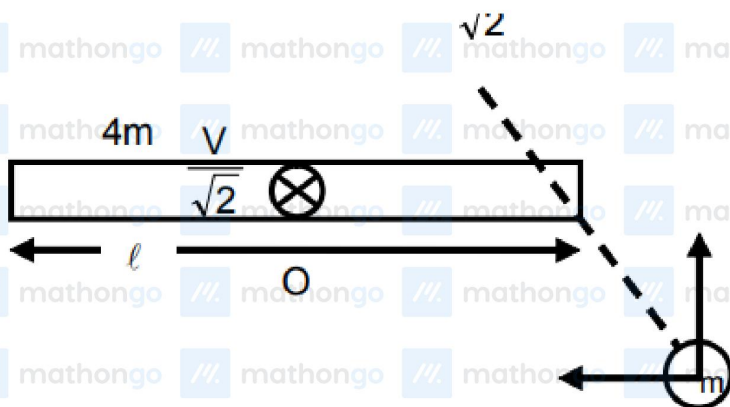
$$\sigma = A + Br$$

Q22

$$L_{oi} = L_{of}$$

$$\frac{mV}{\sqrt{2}} \times \frac{L}{2} = \left[\frac{4mL^2}{12} + \frac{mL^2}{4} \right] \times \omega$$

$$\omega = \frac{6V}{7\sqrt{2}L} = \frac{3\sqrt{2}V}{7L}$$



Q23

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Rotation

Hints & Solutions

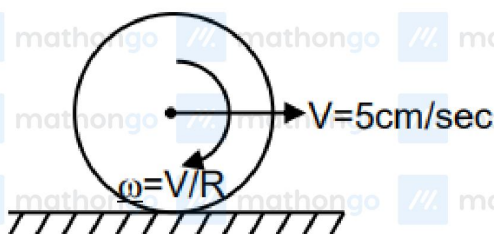
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Physics

K.E. of the sphere = Translational K.E + Rotational K.E.

$$= \frac{1}{2}mv^2 \left(1 + \frac{K^2}{R^2}\right) \quad K = \text{Radius of gyration}$$

$$\frac{1}{2} \times \frac{1}{2} \times \left(\frac{5}{100}\right)^2 \left(1 + \frac{2}{5}\right) \times 10^{-4} J$$



Q24

$$\text{M.I about } P = 3 \left[\frac{2}{5} M \left(\frac{d}{2}\right)^2 + M \left(\frac{d}{\sqrt{3}}\right)^2 \right] = \frac{13}{10} M d^2$$

$$\text{M.I about } B = 2 \left[\frac{2}{5} M \left(\frac{d}{2}\right)^2 + M(d)^2 \right] + \frac{2}{5} M \left(\frac{d}{2}\right)^2$$

$$= \frac{23}{10} M d^2$$

$$\text{Now ratio} = \frac{13}{23}$$

Q25

Rotation

Hints & Solutions

JEE Main 2020 Chapterwise

Physics

$$mgh = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$v = \omega R \text{ (no slipping)}$$

$$mgh = \frac{1}{2}m\omega^2 R^2 + \frac{1}{2} \frac{mR^2}{2} \omega^2$$

$$mgh = \frac{3}{4}m\omega^2 R^2$$

$$\omega = \sqrt{\frac{4gh}{3R^2}} = \frac{1}{R} \sqrt{\frac{4gh}{3}}$$